

# **EcoGrid Pavement Project Report**

## **Executive Summary**

EcoGrid Pavement is a sustainable pavement concept that converts footsteps into electrical energy while incorporating rainwater drainage and natural filtration. It is designed for schools and public walkways.

## **Problem Statement**

Many walkways waste the energy produced by footsteps and contribute to poor stormwater management.

## **Objectives**

Harvest footstep energy, improve safety with lighting, manage rainwater, and promote clean energy awareness.

## **System Design**

The design includes a bamboo composite walking surface, force distribution plate, springs, kinetic generator, rectifier, charge controller, rechargeable battery, LED lighting, drainage channels, filtration layers, and water storage.

## **Expected Benefits**

Renewable energy generation, improved campus safety, environmental sustainability, educational value, and reduced energy waste.

## **Materials**

Bamboo composite, springs, generator, bridge rectifier, charge controller, rechargeable battery, LEDs, sand, gravel, activated carbon, waterproof casing.

## **Conclusion**

EcoGrid Pavement demonstrates how everyday human movement can contribute to clean energy and sustainable infrastructure.